



Christian Doppler Laboratory MEVSS
Monitoring and Evolution of
Very-Large-Scale Software Systems

Position Paper:

Evolving Systems of Systems: Industrial Challenges and Research Perspectives

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<http://mevss.jku.at>

Siemens VAI - Industrial Use Case

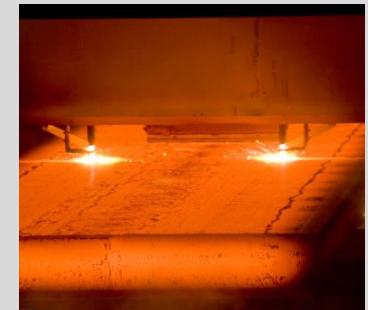
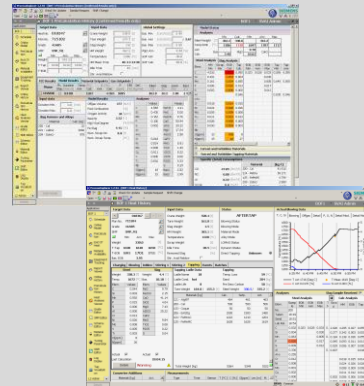
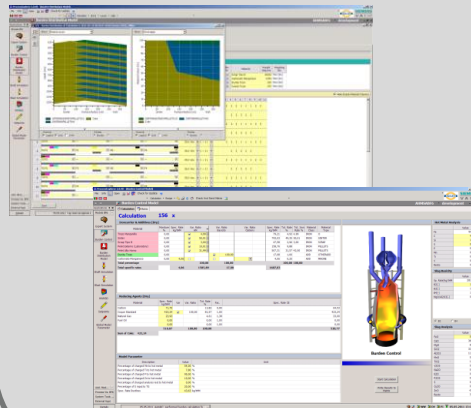
Steel Plant Automation

- Ironmaking

- Steelmaking

- Continuous Casting

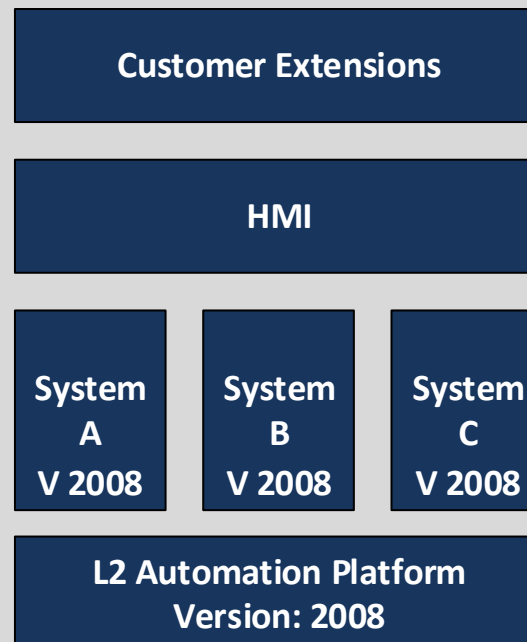
L2 Automation SoS



Selective Upgrade/Evolution

The past: **Many green field plants with complete Level 2 automation systems** (full scope of delivery)

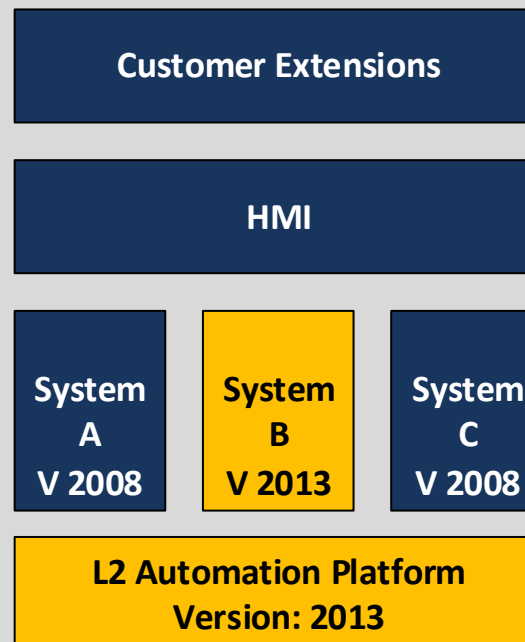
Today: **Upgrade and Service Business**



Selective Upgrade/Evolution

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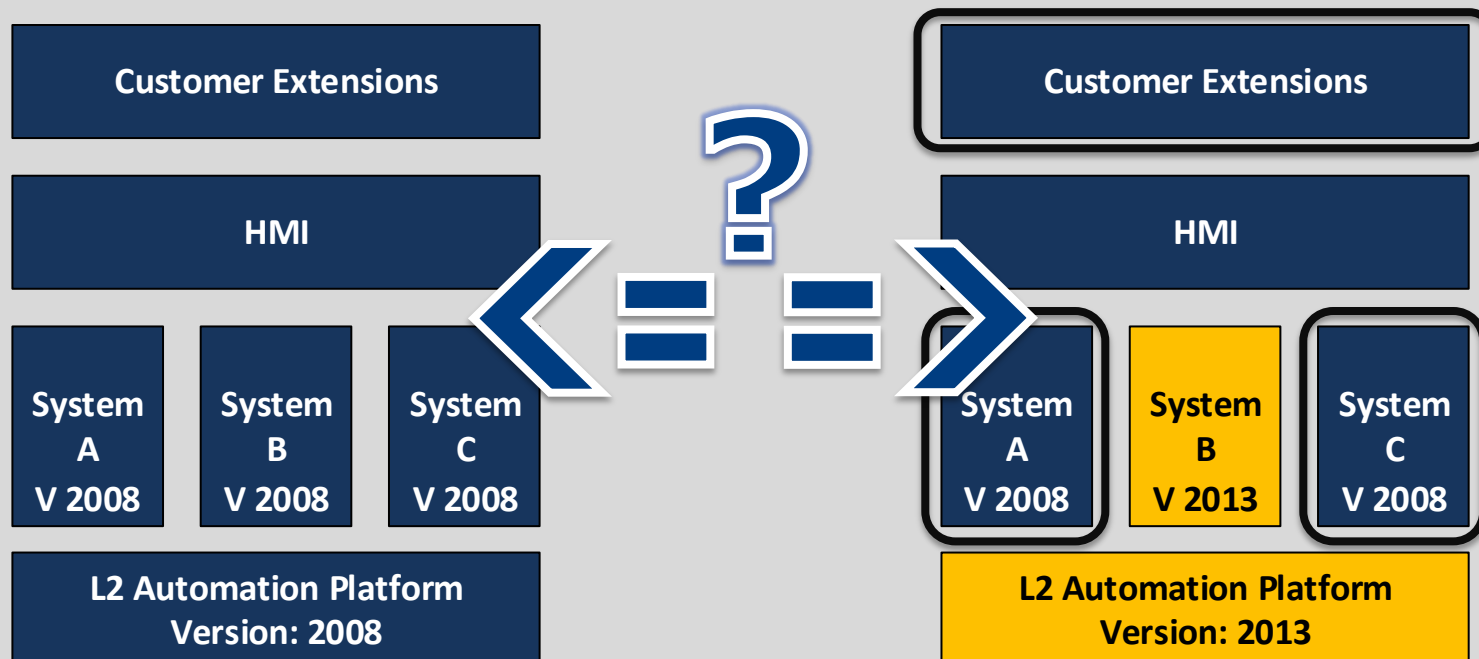
Today: **Upgrade and Service Business**



Selective Upgrade/Evolution

The past: **Many green field plants with complete Level 2 automation systems** (full scope of delivery)

Today: **Upgrade compatible Runtime behavior**



Requirements Monitoring

- Maiden 2013 : *Monitoring our requirements.*
- Robinson 2010: *A Roadmap for Comprehensive Requirements Modeling.*
- Sawyer et al. 2010: *Requirements-Aware Systems*

System Monitoring

- van Hoorn et al. 2012: *Kieker: A Framework for Application Performance Monitoring and Dynamic Software Analysis.*
- Cleland-Huang et al. 2003: *Event-based traceability for managing evolutionary change.*

Variability Management

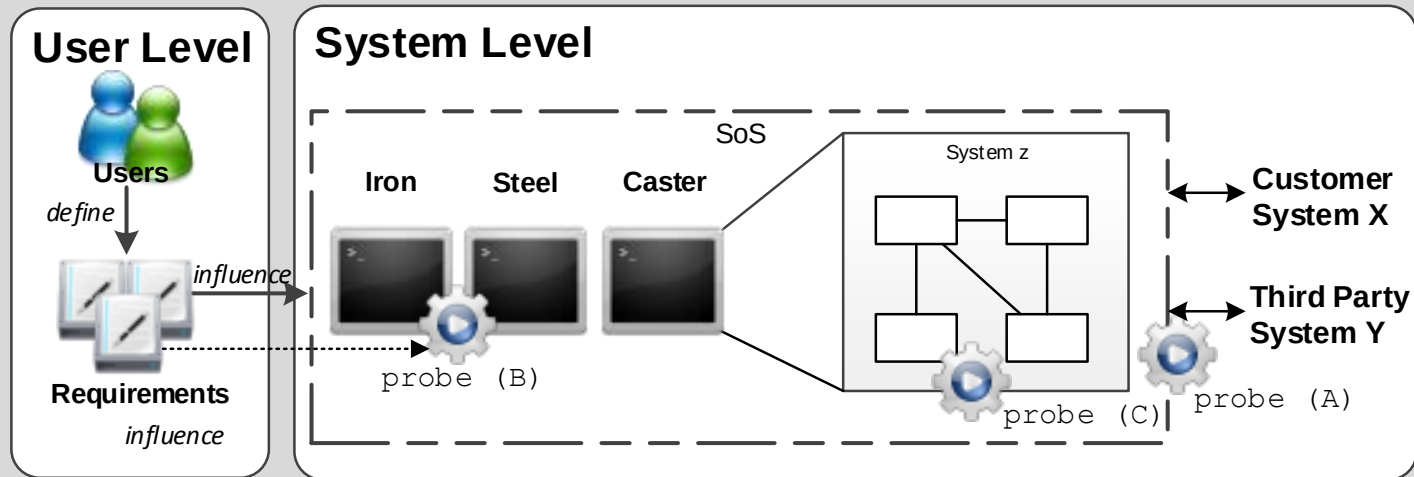
- Czarnecki et al. 2012: *Cool Features and Tough Decisions: A Comparison of Variability Modeling Approaches*



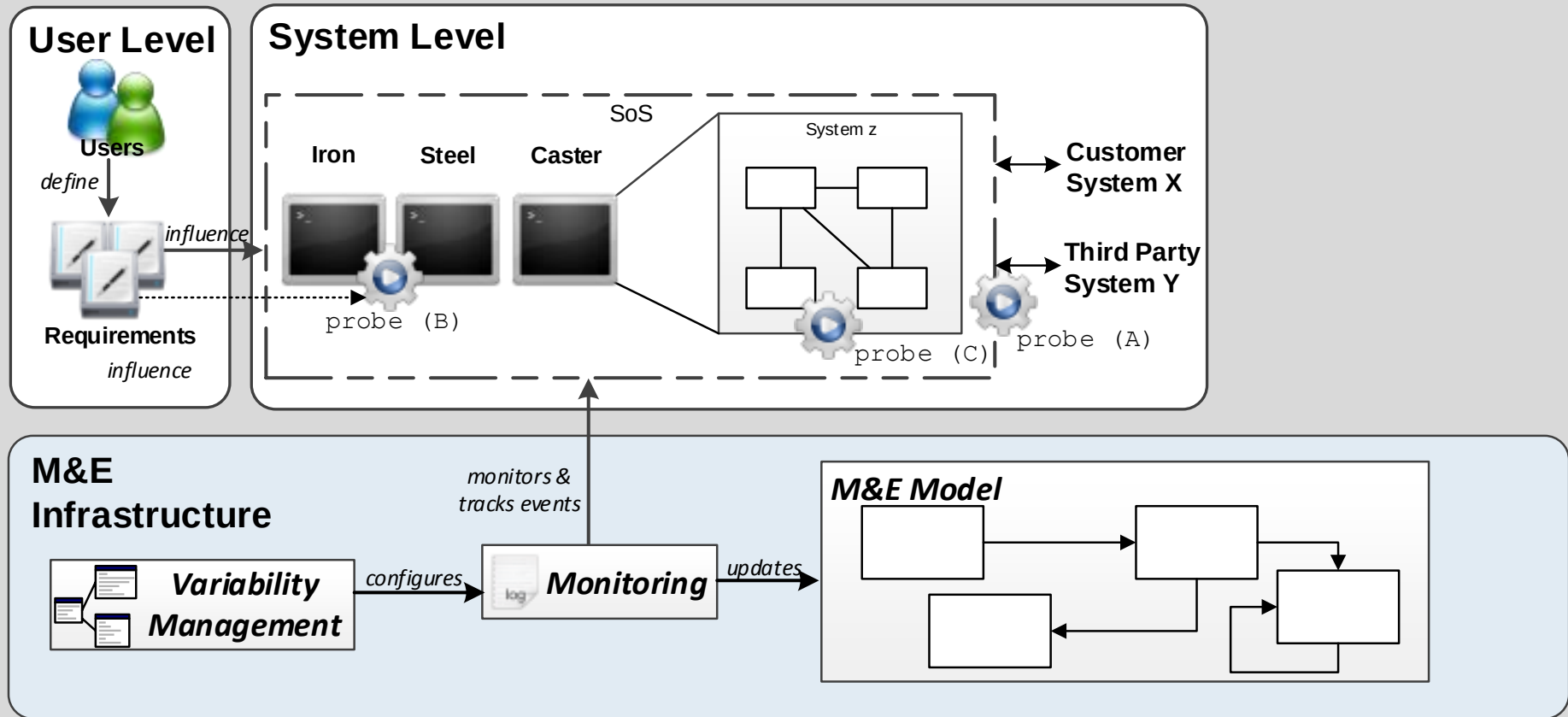
Shortcomings

Focus on single system only
Scope only on one application area

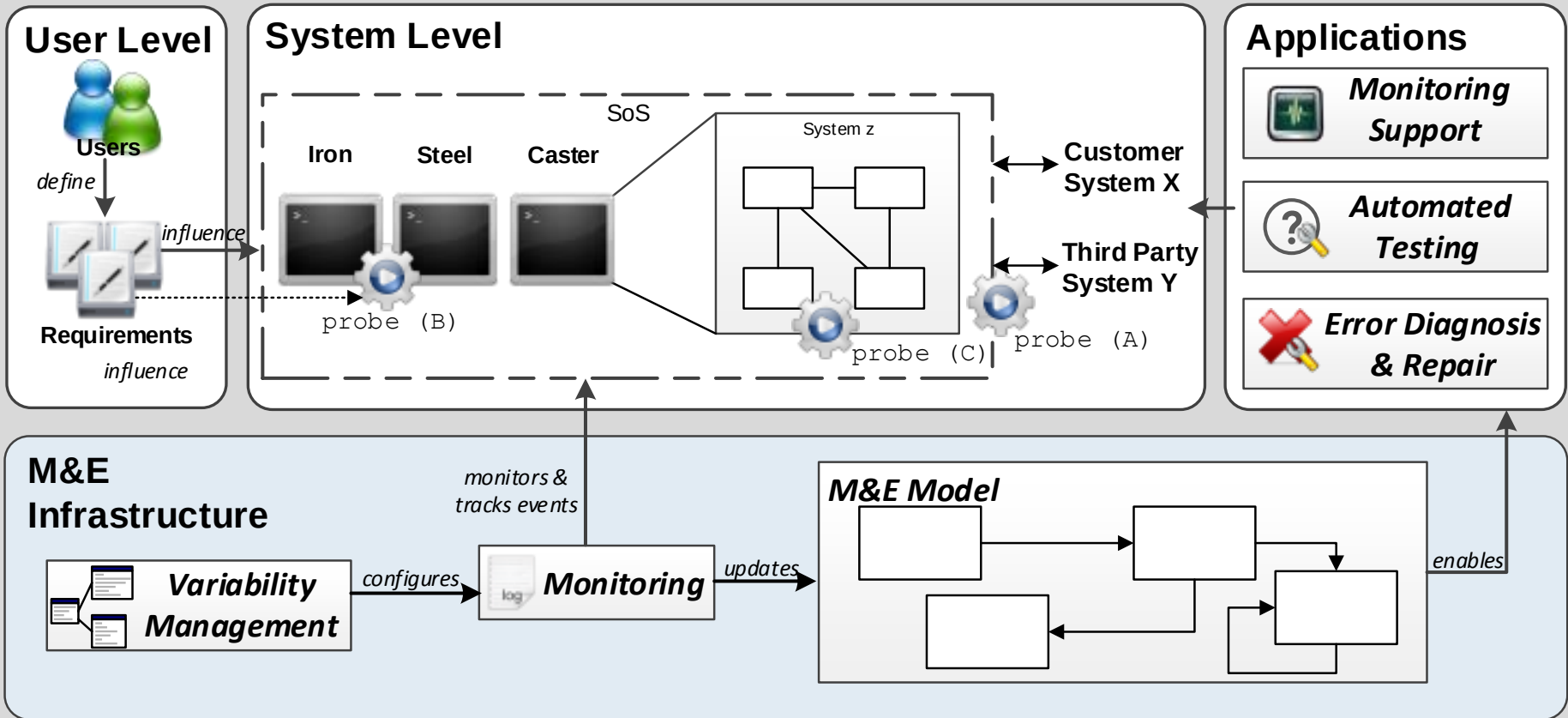
SoS Monitoring & Evolution Support



SoS Monitoring & Evolution Support

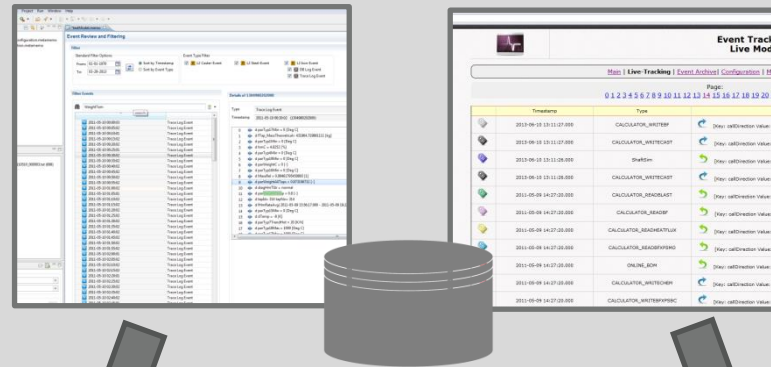


SoS Monitoring & Evolution Support

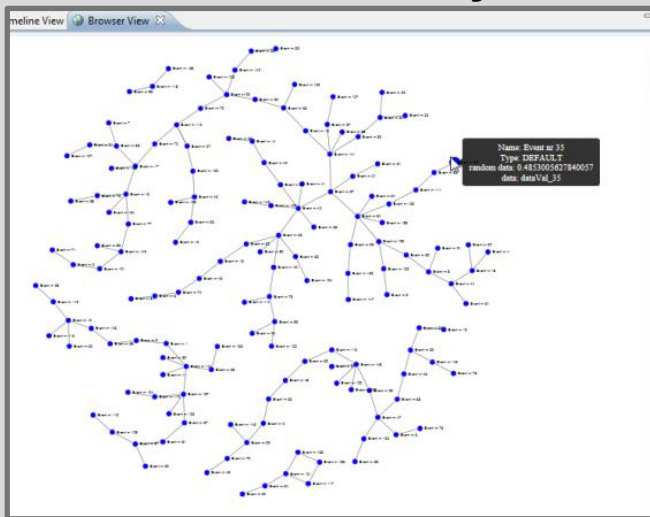


Tool Support Prototypes

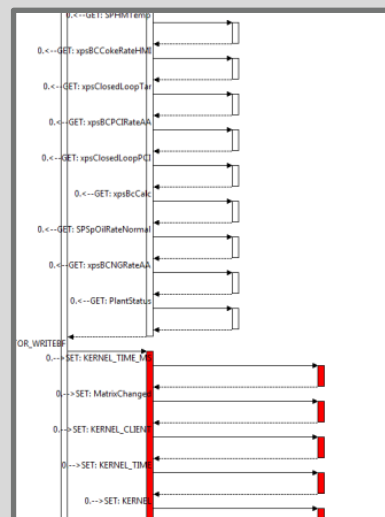
Event Viewer



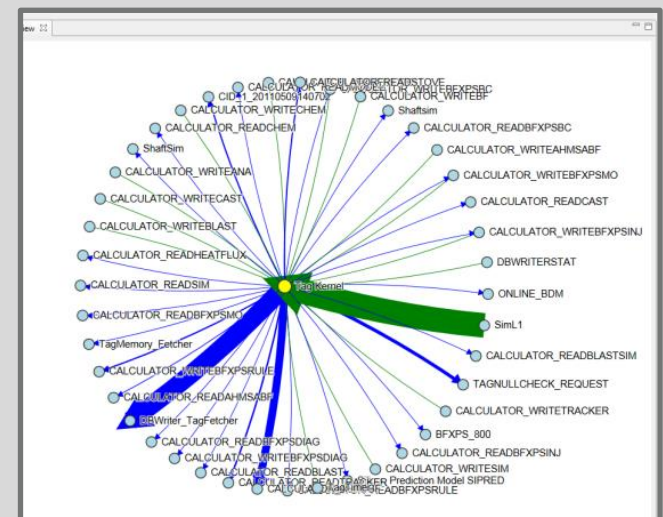
Call Hierarchy



Call Order



Event Flow



Monitoring & Evolution

- Concept for a model-based Monitoring & Evolution Infrastructure
- Development of research prototypes
- As-Is assessment with workshops & interviews

Open Issues

- “Non-intrusive” instrumentation of diverse systems
- “Big Data”/Persistence
- Performance/Life-tracking

Next Steps

- Prototype refinement
- Further workshops & interviews with system experts
- Feedback and evaluation from developers & system users

Thank You!



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<http://www.jku.at>